

Business Impact of Process Improvements Workshop

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ABSTRACT

The topic of the workshop “Business Impact of Process Improvements” is achieving tangible and sustainable business impact from process improvements. Focus is on approaches that are both practical and quantifiable. The kinds of process improvements addressed include, but are not limited to, the introduction of iterative, agile or “lean” approaches, improved requirements engineering, risk management, usage of process improvement reference frameworks, and improved quality control and assurance. Business impact should be illustrated through either increasing the value of what is delivered to the customer or reducing the cost (for example by reduction of rework effort). The audience will share experiences how to reliably set-up, measure and achieve business-oriented process improvements in order to increase return on investment in software engineering.

Categories and Subject Descriptors

D.2.8 [Software Engineering]: Metrics - *Complexity measures, Performance measures, Process metrics, Product metrics*
D.2.9. [Software Engineering]: Management - *Cost estimation, Time estimation, Life cycle, Productivity, Programming teams, Software process models (e.g., CMM, ISO, PSP), Software quality assurance (SQA)*

General Terms

Process Management, Measurement, Performance, Economics, Human Factors

Keywords

Process improvement, business benefits, business impact, cost-benefit analysis, agile and iterative development, effort estimation, QFD, ROI, cause and effect chains, performance, economics

1. SCOPE

Software process improvement is key to align engineering with business needs. A lot of effort and resources is put on process improvement, but not so often the effects are visible in the business scorecards. A majority of all improvement and reengineering programs fail to deliver value. The topic of this workshop is achieving tangible and sustainable business impact from process improvements.

The goal of the “Business Impact of Process Improvement” (BIPI) workshop is to establish a forum where practical and quantifiable value in process improvement is reviewed and consolidated to advance software engineering improvement programs.

Too often, there is too little connection to the impact for the business and in such a way that the recommended approaches are able to be applied in practice.

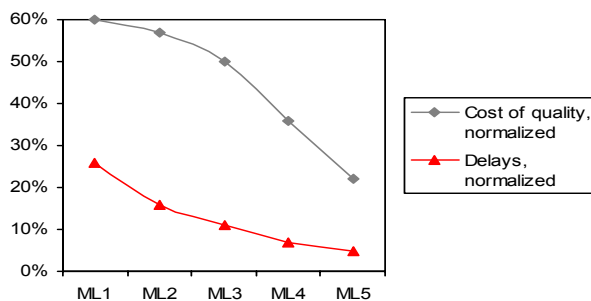
We explicitly strive to serve as an appropriate forum for bridging the academic, research, business, and industrial practice communities.

BIPI-2008 looks to improvement dimensions such as:

- State of the practice engineering methodology, such as model-driven development
- Agile, lean or flexible engineering and management practices
- Incremental delivery
- Iterative development
- Improved communication and interaction with customers and end users
- Strengthened requirements engineering
- Using a maturity framework such as CMMI or SPICE
- Quantitative and statistical management, such as Six Sigma
- Better innovation management
- World-wide collaboration with distributed engineering teams, suppliers, partners, and clients
- Product life-cycle management
- Quality control techniques.

Assuming, as a basis that an organization has an at least repeatable process and has a basic measurement system in place, then the organization should be able to achieve and measure the business impacts through either increasing the value of what is delivered to the customer (such as better selection of requirements) or reducing the cost to deliver (such as reducing rework). Improved effort estimation, higher team motivation, shorter time-to-market are further benefits that impact the business.

Different empirical studies show that process improvement has profound impact on business – if done in a objective-driven approach. One important contribution of process improvement to business success is highlighted by the periodic evaluations of project success by the Standish Group [1]. While in 1994 only 16% of all IT projects could be considered successful, for those projects finished in 2006 it was already a success rate of 35%. The average cost overrun in 1994 was more than 100%, while today it is less than 40%. Within 12 years a strong momentum is visible which was initiated and driven by a much stronger process focus than ever before. Process improvement increases project success rate. The following figure summarizes two longitudinal studies done at Motorola and Alcatel-Lucent during the past years [1]. Project performance was measured along the respective maturity level following the CMMI (horizontal axis) with the two dimensions of normalized cost of quality and normalized delays at project end (vertical axis). With increasing process maturity – which is an investment in process improvement – there is a tangible business impact in terms of reduced cost of quality and less delays. With such data being available from different organizations it is fair to state that – if done well – process improvement has a strong business impact with sustainable ROI [2].



To sum up our experiences, process improvement must always be driven from clear business objectives. Never start a process improvement activity with unclear objectives, or by simply saying that a certain CMMI maturity level must be achieved. It will fail and deliver mediocre results. Improve project performance based on specified criteria of predictability, quality and productivity. These criteria must be derived from your business goals and should be approved or reviewed with your upper management.

The participants will share experiences how to reliably set-up, measure and achieve business-oriented process improvements in order to increase return on investment in software engineering.

2. PRESENTATIONS

The workshop format is a mix of presentations and goal-oriented discussions. It will include a keynote from Hakan Erdogmus (NRC) and the following presentations:

- “Capturing business benefits from process improvement: Four fallacies and what to do about them” by Paul Bannerman from NICTA (Australia)
- “Strategic alignment of software process improvement programs using QFD” by André Luis Becker, Rafael Prikladnicki, and Jorge Luis Nicolas Audy from PUCRS, (Brazil)

- “A framework for software project estimation based on COSMIC, DSM and rework characterization” by Sharrareh Afsharian from University of L’Aquila and Ericsson Lab, Marco Giacomobono from Ericsson Lab, and Paola Inverardi from University of L’Aquila (Italy)
- “Using opinion polls to help measure business impact in agile development” by Harald Klein and Sabine Canditt from Siemens AG (Germany)
- “Towards a Model for Cost-Benefit-Analysis of Quality Assurance in the Automotive E/E Development” by Nicolas F. Porta from Daimler AG (Germany)

3. ORGANIZERS AND COMMITTEE

Dr. Frances Paulisch is a software “best practice” evangelist at the Corporate Technology division of Siemens AG in Munich, Germany. She is responsible for the “Siemens Software Initiative” which enables the sharing of best practices related to software and quality throughout the Siemens enterprise. Three of the main current areas of interest are measurement of software-based systems, global software engineering, and requirements engineering. She received her doctorate at the University of Karlsruhe in Germany. She is on the industrial advisory board of the IEEE Software magazine. Furthermore, she is a member of the IEEE and the German “Gesellschaft für Informatik”.

Dr. Christof Ebert is managing director at Vector Consulting Services, where he is implementing process improvement activities with business impact world-wide. Prior to that, he held engineering and management positions at Alcatel for more than a decade, most recently as Director of Engineering with global responsibility for software technologies, engineering processes and tools. A senior member and distinguished visitor of IEEE, Dr. Ebert authored several books. He currently chairs the IEEE Intl. Conf. on Global Software Engineering (ICGSE) steering board. He is engaged in standardization activities, lectures at University of Stuttgart and serves on editorial boards of several scientific journals and as a frequent keynote speaker at conferences.

The Program Committee consists of:

- Rob Austin, Harvard Business School, USA
- Ton Dekkers, Shell, Netherlands
- Hakan Erdogmus, NRC, Canada
- Harald Klein, Siemens AG, Germany
- Marek Leszak, Alcatel-Lucent, Germany
- Pat O’Toole, PACT, USA
- Roman Pichler, Pichler Consulting, UK
- Wolfgang Strigel, independent consultant, Germany
- Rini Van Solingen, TU Delft, Netherlands

4. FURTHER INFORMATION

All further information on this workshop can be found at the URL: <http://www.bipi-ws.org>

[1] Ebert, C., and R.Dumke: Software Measurement. Springer, Heidelberg, New York, 2007.

[2] Performance Results of CMMI®-Based Process Improvement. CMU/SEI-2006-TR-004. August 2006